

FCFS CPU Scheduling

Aim

To write and execute a Python program to implement the First-Come, First-Served (FCFS) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1                                fcfs.py *
def find_wt(process, n, bt, at):
    wt = [0] * n
    service_time = [0] * n
    service_time[0] = at[0]

    for i in range(1, n):
        service_time[i] = service_time[i - 1] + bt[i - 1]
        wt[i] = service_time[i] - at[i]

        if wt[i] < 0:
            wt[i] = 0
            service_time[i] = at[i]

    return wt

def find_TAT(n, bt, wt):
    tat = [0] * n

    for i in range(n):
        tat[i] = bt[i] + wt[i]

    return tat

def find_avg_time(process, n, bt, at):
    proc_info = sorted(
        zip(process, at, bt),
        key=lambda x: x[1]
    )

    process, at, bt = zip(*proc_info)

    wt = find_wt(process, n, bt, at)
    tat = find_TAT(n, bt, wt)
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

^C Location
^/ Go To Line

Right Ctrl

```

total_wt = sum(wt)
total_tat = sum(tat)

print("Execution order:", " -> ".join(map(str, process)))

print("\nProcess\tArrival\tBurst\tWaiting\tTurnaround")

Settings i in range(n):
    print(
        f"{process[i]}\t{at[i]}\t{bt[i]}"
        f"\t{wt[i]}\t{tat[i]}"
    )

print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
print(f"Average Turnaround Time = {total_tat / n:.2f}")

if __name__ == "__main__":
    process = [1, 2, 3, 4]
    arrival_time = [0, 1, 2, 3]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time,
        arrival_time
    )

```

^{^G} Help ^{^O} Write Out ^{^F} Where Is ^{^K} Cut ^{^T} Execute ^{^C} Location
^{^X} Exit ^{^R} Read File ^{^_\} Replace ^{^U} Paste ^{^J} Justify ^{^/} Go To Line

Output :

Output for the given process, arrival-time, and burst-time values:

```

python3: can't open file '/home/aayush/fcfs.py': [Errno 2] No such file or directory
aayush@ubuntu:~$ nano fcfs.py
aayush@ubuntu:~$ python3 fcfs.py
Execution order: 1 -> 2 -> 3 -> 4

Process Arrival Burst   Waiting Turnaround
1         0         5         0         5
2         1         3         4         7
3         2         8         6        14
4         3         6        13        19

Average Waiting Time = 5.75
Average Turnaround Time = 11.25
aayush@ubuntu:~$

```

Result

The FCFS scheduling program was executed. The average waiting time is 5.75 time units and the average turnaround time is 11.25 time units.

SJF CPU Scheduling

Aim

To write and execute a Python program to implement the Shortest Job First (SJF) CPU scheduling algorithm and calculate the waiting time and turnaround time for each process, along with their averages.

Program

```
GNU nano 8.7.1 sjf.py *
def find_waiting_time(process, n, bt, wt):
    wt[0] = 0

    for i in range(1, n):
        wt[i] = bt[i - 1] + wt[i - 1]

def find_turnaround_time(n, bt, wt, tat):
    for i in range(n):
        tat[i] = bt[i] + wt[i]

def find_avg_time(process, n, bt):
    wt = [0] * n
    tat = [0] * n

    # Sort processes according to burst time
    for i in range(n):
        for j in range(i + 1, n):
            if bt[i] > bt[j]:
                bt[i], bt[j] = bt[j], bt[i]
                process[i], process[j] = process[j], process[i]

    find_waiting_time(process, n, bt, wt)
    find_turnaround_time(n, bt, wt, tat)

    total_wt = sum(wt)
    total_tat = sum(tat)

    print("SJF Execution Order:")
    print(" -> ".join(map(str, process)))

    print("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time")

    for i in range(n):
        print(
            f"{process[i]}\t{bt[i]}\t\t"
            f"{wt[i]}\t\t{tat[i]}"
        )

    print(f"\nAverage Waiting Time = {total_wt / n:.2f}")
    print(f"Average Turnaround Time = {total_tat / n:.2f}")

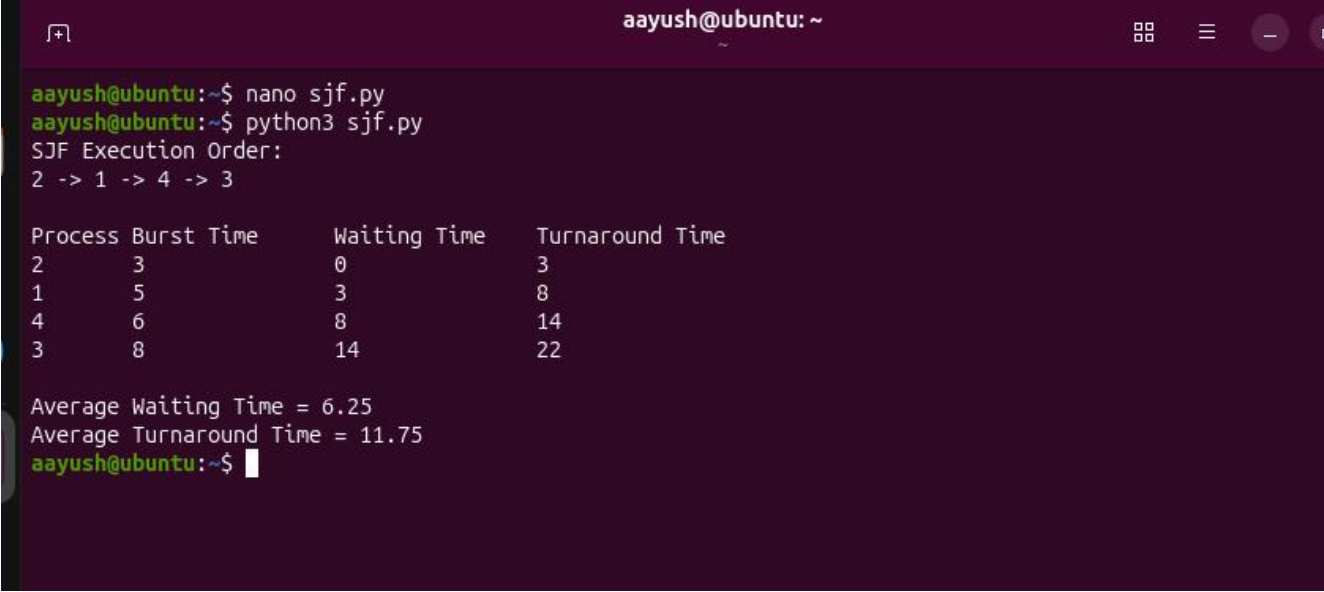
if __name__ == "__main__":
    process = [1, 2, 3, 4]
    burst_time = [5, 3, 8, 6]

    find_avg_time(
        process,
        len(process),
        burst_time
    )
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Right Ct

Output Screenshot

Output for the given burst-time values:

A terminal window with a dark purple background. The title bar shows 'aayush@ubuntu: ~'. The terminal contains the following text:

```
aayush@ubuntu:~$ nano sjf.py
aayush@ubuntu:~$ python3 sjf.py
SJF Execution Order:
2 -> 1 -> 4 -> 3

Process Burst Time    Waiting Time    Turnaround Time
2         3           0              3
1         5           3              8
4         6           8             14
3         8          14             22

Average Waiting Time = 6.25
Average Turnaround Time = 11.75
aayush@ubuntu:~$
```

Result

The SJF scheduling program was executed successfully. The average waiting time is 6.25 time units and the average turnaround time is 11.75 time units.